



**CS344: Operating
Systems Lab Quiz,
October 23, 2024
Set - C**

Indian Institute of Technology Guwahati

Duration: 40 Minutes

Total : 20 Marks

Name: _____

Roll No.: _____

Student Signature: _____

Invigilator Signature: _____

(10 Questions)

Part - A

[10 Marks]

> Please mention only the option you select inside the box in capital letters.

Q1. When implementing lazy memory allocation in xv6, how does the kernel trick a process into believing it has the requested memory?

- a>** The kernel swaps out pages to free up memory for the process.
- b>** The kernel marks the requested memory as inaccessible until the process accesses it.
- c>** The kernel immediately allocates physical memory but hides this from the process.
- d>** The sbrk() system call increases the process's virtual memory size without allocating physical memory.

Q2. What is the significance of the address 0x7c00 during the boot process?

- a>** It is where the BIOS is stored.
- b>** It is the starting address of the kernel.
- c>** It is the location where the boot sector is loaded into memory.
- d>** It is the default address for user programs.

Q3. Which of the following steps is NOT required when implementing a `getNumProc()` system call in the xv6 operating system?

- a> Modifying the `proc` structure to track the number of active processes.
- b> Implementing the logic in the kernel to count active processes across all states.
- c> Modifying the system call table to include `getNumProc()`.
- d> Writing a user-level program to invoke `getNumProc()`.

Q4. Which of the following is a key feature of paging in operating systems?

- a> Memory is allocated contiguously for all processes.
- b> Processes are limited to using only available physical memory.
- c> Memory can be divided into fixed-size pages that are swapped in and out of physical memory.
- d> Memory can be divided into fixed-size pages that are swapped in and out of physical memory.

Q5. What is the primary purpose of the `panic()` function in the xv6 operating system kernel?

- a> To handle system errors by trying to recover and continue execution.
- b> To log an error message and halt the system immediately.
- c> To reset the kernel and restart the operating system.
- d> To send a warning to the user space and allow the process to terminate gracefully.

Q6. What does the sys_draw system call do when the provided buffer is too small to hold the ASCII art image?

- a> It retries with a larger buffer.
- b> It returns a positive value.
- c> It crashes the system.
- d> It returns a negative value.

Q7. When does a PC switch from 16-bit real mode to 32-bit protected mode during booting?

- a> After the A20 address line is configured.
- b> Once the user-level program starts executing.
- c> As soon as the BIOS initializes.
- d> When the boot loader reads the kernel.

Q8. What happens if the sys_draw system call fails because the buffer is too small?

- a> The system crashes.
- b> The system call returns the number of bytes copied.
- c> The system call will retry with a larger buffer.
- d> The system call returns a negative value.

Q9. What will happen if a user forgets to provide a co command-line argument to the sleep program?

- a> The program will print an error message.
- b> The program will pause indefinitely.
- c> The program will use a default value of 10 ticks.
- d> The program will terminate without any output.

Q10. Which of the following steps is essential when adding a new system call to xv6?

- a>** Add the system call to kalloc.c to allocate memory for the process.
- b>** Add the system call to console.c to enable debugging output.
- c>** Modify trap.c to handle the new system call.
- d>** Modify syscall.h to add a new system call number.

10 Questions

Part-B

[10 Marks]

Q1. Which of the following accurately describes the role of the ZOMBIE state in xv6?

- a>** A process in the ZOMBIE state is still actively running but waiting for resources.
- b>** A process in the ZOMBIE state has terminated but its parent has not yet called wait() to collect its exit status.
- c>** A process in the ZOMBIE state is suspended and can be resumed at a later time.
- d>** A process in the ZOMBIE state is paused due to an error and can only be restarted after debugging.

Q2. Which kernel organization allows most of the operating system to run in user mode while minimizing the code running in kernel mode?

- a>** Monolithic kernel.
- b>** Microkernel.
- c>** Hybrid kernel.
- d>** Exokernel.

Q3. In xv6, which combination of page table entry (PTE) flags would allow a process to execute code but prevent it from writing to memory, and how does this contribute to system stability?

- a> PTE_P | PTE_U | PTE_W.
- b> PTE_P | PTE_U | ~PTE_W.
- c> PTE_P | PTE_W.
- d> PTE_P | PTE_U.

Q4. In real mode (the mode that PC starts off in, before xv6 OS is booted), how are addresses calculated from CS (segment) and IP (offset) registers ?

- a> (segment x offset)
+ 100.
- b> (segment | 0xFF00) + (offset & 0xF0A0).
- c> (16 x segment) + offset.
- d> segment + (offset << 4).

Q5. What happens when an application in user mode attempts to execute a privileged instruction?

- a> The processor switches to kernel mode for handling.
- b> The processor immediately halts the system.
- c> The processor executes the instruction normally.
- d> The processor ignores the instruction.

Q6. Which fields are to be added to the proc struct for performance evaluation?

- a> ppid, pid, name.
- b> ctime, stime, retime, runtime.
- c> start_time, end_time, total_time.
- d> process_id, user_id, status.

Q7. What instruction does the BIOS execute first after the PC is powered on ?

- a> mov instruction to initialize registers.
- b> jmp instruction to jump to a different location in BIOS.
- c> call instruction to invoke a subroutine.
- d> nop instruction to skip to the next step.

Q8. When adding the settickets() system call, which of the following steps ensures that children processes inherit the correct number of tickets after a fork() operation?

- a> Modifying the fork() system call to copy the parent's ticket count to the child process.
- b> Automatically setting child processes to default tickets (10) without copying the parent's tickets.
- c> Making changes in sysproc.c to call settickets() after every fork().
- d> The child process independently requests tickets through settickets() during creation.

Q9. How does the bootloader know where the kernel's first instruction is located?

- a> It calculated the address based on memory size.
- b> It is hard-coded in the bootloader.
- c> It is specified in the BIOS settings.
- d> It reads the `e_entry` field from the ELF header.

Q10. What is the default scheduling policy used in the xv6 operating system ?

- a> Priority-based scheduling.
- b> First-Come, First-Served (FCFS).
- c> Round-robin scheduling.
- d> Lottery scheduling.